

Qiuqi Wang, Ph.D., A.S.A.

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Academic Appointment

Assistant Professor 2023–

Education

Ph.D. in Actuarial Science, University of Waterloo – Advisor: Prof. Ruodu Wang 2023
M.Phil. in Mathematics, Hong Kong University of Science and Technology –
Advisor: Prof. Yue Kuen Kwok 2019
B.Sc. in Mathematics and Economics, Hong Kong University of Science and
Technology 2017

Professional Designation

Associate of the Society of Actuaries (A.S.A.) 08/2023

Research Interests

- Actuarial science; quantitative risk management; Operations research; financial engineering;
Non-parametric hypothesis testing; Game theory
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Publications and Manuscripts

Pre-publication Manuscripts

- [1] Jiao, Z., **Wang, Q.** and Zhao, Y. (2026). Comparative e-backtests for general risk measures. Preprint, 2026.
- [2] Bellini, F., Huang, M., **Wang, Q.** and Wang, R. (2026). Lambda Expected Shortfall. Preprint, 2026.
- [3] Chi, Y., Fung, T. C., Peng, L. and **Wang, Q.** (2025). Statistical solutions of optimal multivariate reinsurance under distortion risk measures. Preprint, 2025.

Peer-reviewed Journal Articles

- [1] **Wang, Q.**, Wang, R. and Ziegel, J. (2025). E-backtesting. *Management Science*, available online, 2025.
- [2] Han, X., **Wang, Q.**, Wang, R. and Xia, J. (2025). Cash-subadditive risk measures without quasi-convexity. *Mathematics of Operations Research*, available online, 2025.
- [3] Boonen, T. J., Chen, Y., Han, X. and **Wang, Q.** (2025). Optimal insurance design with Lambda-Value-at-Risk. *European Journal of Operational Research*, **327**(1), 232–246, 2025.

- [4] Aboagye, E., Asimit, V., Fung, T. C., Peng, L. and **Wang, Q.** (2025). A revisit of the optimal excess-of-loss contract. *European Journal of Operational Research*, **322**(1), 341–354, 2025.
- [5] Pesenti, S., **Wang, Q.** and Wang, R. (2024). Optimizing distortion riskmetrics with distributional uncertainty. *Mathematical Programming*, **213**, 51–106, 2024.
- [6] **Wang, Q.**, Wang, R. and Zitikis, R. (2022). Risk measures induced by efficient insurance contracts. *Insurance: Mathematics and Economics*, **103**, 56–65, 2022.
- [7] Embrechts, P., Mao, T., **Wang, Q.** and Wang, R. (2021). Bayes risk, elicibility, and the Expected Shortfall. *Mathematical Finance*, **31**(4), 1190–1217, 2021.
- [8] **Wang, Q.**, Wang, R. and Wei, Y. (2020). Distortion riskmetrics on general spaces. *ASTIN Bulletin*, **50**(3), 827–851, 2020.
- [9] **Wang, Q.** and Kwok, Y. K. (2020). Real option signaling games of debt financing using equity guarantee swaps. *International Journal of Theoretical and Applied Finance*, **23**(5), 2050036, 2020.
- [10] **Wang, Q.** and Kwok, Y. K. (2019). Signaling game models of equity financing under information asymmetry and finite project life. *International Journal of Financial Engineering*, **6**(1), 1950002, 2019.

Dissertations

- [1] Wang, Q. (2023). Characterizing, optimizing and backtesting metrics of risk. *Ph.D. Thesis*. University of Waterloo, Canada.
- [2] Wang, Q. (2019). Real option signaling games under asymmetric information and finite option life. *M.Phil. Thesis*. Hong Kong University of Science and Technology, Hong Kong.

Selected Honors and Awards

Pierre Robillard Award	2024
James C. Hickman Scholar Doctoral Stipend	2022–2023
Mathematics Doctoral Prize – 3rd Place	2023

Academic Visits

University of New South Wales (School of Risk & Actuarial Studies) (Sydney, Australia)	2026
University of Melbourne (Department of Economics) (Melbourne, Australia)	2026
Monash University (Department of Econometrics and Business Statistics) (Clayton, Australia)	2026
University of Iowa (Department of Statistics and Actuarial Science) (Iowa City, US)	2026
Michigan State University (Department of Statistics and Probability) (Lansing, US)	2025
University of Waterloo (Department of Statistics and Actuarial Science) (Waterloo, Canada)	2023, 2025
Nankai University (School of Mathematical Sciences) (Tianjin, China)	2023, 2025
Chinese University of Hong Kong-Shenzhen (School of Science and Engineering) (Shenzhen, China)	2024
Chinese Academy of Sciences (Academy of Math and Systems Science) (Beijing, China)	2021, 2023
University of Science and Technology of China (School of Management) (Hefei, China)	2021

Conferences Organized

- 3rd Waterloo Student Conference in Statistics, Actuarial Science and Finance (*Co-chair*), 2022 (Waterloo, Canada)
- 2nd Waterloo Student Conference in Statistics, Actuarial Science and Finance (*Co-chair*), 2021 (Online)

Invited Academic Presentations

- 2nd Midwest Actuarial Research Seminars, 2026 (West Lafayette, US)
- China Conference on Insurance and Risk Management, 2025 (Urumqi, China)
- 3rd Foundations and Applications of Decentralized Risk-Sharing, 2025 (Paris, France)
- Young Talents in Actuarial Science and Quantitative Finance, 2025 (Waterloo, Canada)
- Pierre Robillard Invited Address, SSC Annual Meeting, 2024 (St. John's, Canada)
- Young Data Science Researcher Seminar Zurich, 2022 (Online)
- INFORMS Annual Meeting, 2022 (Indianapolis, US)

Contributed Academic Presentations

- 1st ASTIN Bulletin Conference, 2026 (Zurich, Switzerland)
- 60th Actuarial Research Conference, 2025 (Toronto, Canada)
- 59th Actuarial Research Conference, 2024 (Murfreesboro, US)
- 27th International Congress of Insurance: Mathematics and Economics, 2024 (Chicago, US)
- 57th Actuarial Research Conference, 2022 (Urbana-Champaign, US)
- 25th International Congress on Insurance: Mathematics and Economics, 2022 (Online)
- CORS/INFORMS International Conference, 2022 (Vancouver, Canada)
- 56th Actuarial Research Conference, 2021 (Online)
- 24th International Congress on Insurance: Mathematics and Economics, 2021 (Online)
- 5th PKU-NUS Annual International Conference on Quantitative Finance and Economics, 2021 (Online)
- 5th Asian Quantitative Finance Seminar, 2020 (Online)
- 1st Waterloo Student Conference in Statistics, Actuarial Science and Finance, 2019 (Waterloo, Canada)
- 23rd Annual International Real Options Conference, 2019 (London, UK)

Teaching Experience

Instructor – Georgia State University

RMI 8300 - Predictive Risk Modeling	Fall 2024, 2025
AS 3230/8230 - Financial Mathematics	Spring 2024; Fall 2024, 2025

Instructor – University of Waterloo

ACTSC 371 - Introduction to Investment	Winter 2023
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Teaching Assistant – University of Waterloo

ACTSC 363 - Casualty and Health Insurance Mathematics 1	Fall 2022
ACTSC 964 - Quantitative Risk Management	Spring 2022
ACTSC 446/846 - Mathematics of Financial Markets	Winter 2022
MATBUS 472 - Risk Management	Winter 2021–2022
MATBUS 470 - Derivatives	Fall 2019–2022; Winter 2021
MATBUS 471 - Fixed Income Securities	Fall 2019–2021; Winter 2020
ACTSC 431 - Loss Model I / Casualty and Health Insurance Mathematics 2	Spring 2020, 2021
ACTSC 445/845 - Quantitative Enterprise Risk Management	Spring 2020
ACTSC 231 - Introductory Financial Mathematics	Winter 2020

Teaching Assistant – HKUST

MATH 4321 - Game Theory (with tutorials)	Spring 2018, 2019
MATH 4511 - Quantitative Methods for Fixed Income Derivatives (with tutorials)	Fall 2018

Peer-review Service

- Actuarial: ASTIN Bulletin, Insurance: Mathematics and Economics, Annals of Actuarial Science, North American Actuarial Journal, European Actuarial Journal
 - Finance: Quantitative Finance, Mathematical Finance, Finance & Stochastics
 - OR: Mathematics of Operations Research
 - Probability: Statistics & Probability Letters
 - Economics: Economic Modelling
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